Java Syntax Study Workbook

Color workbook for learning Java syntax through mini-problems, short sessions, and repeatable review. Built for low-friction ADHD study.

10 min learn | 15 min mini-problems | 10 min old review | 5 min mistake log

STUDY RULE

One topic. Tiny reps. Stop while the session still feels clean.

HOW TO USE IT

- Pick one syntax topic only.
- Copy one example with notes.
- Rewrite it from memory.
- Solve 2 mini-problems.
- Redo 1 old mini-problem.
- Write what you forgot before stopping.

MOVE ON RULE

- You can write the syntax without notes.
- You solved at least 4 of 5 mini-problems cleanly.
- You know when to use it in a DSA problem.

30-DAY ROADMAP

Days 1-5
Variables, arithmetic, if/else, for, while, methods.

Days 6-10
Arrays, strings, review and speed reps.

Days 11-15
Nested loops, brute force thinking, ArrayList.

Days 16-20
HashSet, HashMap, frequency patterns, Two Sum.

Days 21-25
Deque as stack and queue, PriorityQueue.

Days 26-30
Recursion, 2D arrays, sorting, binary search.

WEEKLY REVIEW TARGETS

WEEK	PRIMARY GOAL	REVIEW FOCUS
1	Write basic syntax without looking.	Redo loops and conditionals from memory.
2	Feel comfortable with arrays and strings.	Spot indexing mistakes fast.
3	Use lists and nested loops cleanly.	Separate syntax mistakes from logic mistakes.
4	Map simple DSA problems to Set or Map.	Practice frequency counting.
5	Use stack, queue, and heap syntax naturally.	Review push/pop, offer/poll, offer/poll.

Mini-Problem Bank

Solve tiny problems until the syntax starts to feel automatic. Keep them short. Repeat old ones often.

CORE BASICS

- Print your age.
- Add two numbers.
- Rectangle area.
- Swap two values.
- Minutes to seconds.

IF / ELSE

- Even or odd.
- Positive, negative, or zero.
- Max of two numbers.
- Is vowel.
- Score to grade.

LOOPS

- Print 1 to 10.
- Print 10 to 1.
- Sum 1 to n.
- Multiplication table.
- Count digits.

METHODS

- `add(a, b)`
- `isEven(n)`
- `max(a, b)`
- `square(n)`
- `countDigits(n)`

ARRAYS

- Print all elements.
- Find sum.
- Find maximum.
- Find target index.
- Count evens.

STRINGS

- Count vowels.
- Reverse a string.
- Check palindrome.
- Count uppercase letters.
- Count char frequency.

COLLECTIONS MINI-PROBLEMS

SYNTAX	MINI-PROBLEMS
ArrayList	Add 5 values, sum the list, remove evens, reverse print.
HashSet	Contains duplicate, count distinct numbers, common elements of two arrays.
HashMap	Count frequencies, first repeated number, first unique character, Two Sum.
Deque stack	Reverse string, valid parentheses, remove adjacent duplicates.
Deque queue	Queue simulation, print queue order, BFS on tiny graph.
PriorityQueue	K smallest, K largest, repeated min extraction.

QUICK SYNTAX REFERENCES

```
List<Integer> list = new ArrayList<>();
list.add(10);
int x = list.get(0);

Set<Integer> set = new HashSet<>();
set.add(5);
boolean ok = set.contains(5);
```

```
Map<Integer, Integer> map = new HashMap<>();
map.put(3, 1);
int c = map.getOrDefault(3, 0);

Deque<Integer> stack = new ArrayDeque<>();
stack.push(4);
int top = stack.pop();
```

Daily Practice Pages

Use these pages repeatedly. Print multiple copies if useful.

DAILY PLAN

TOPIC

SYNTAX TO LEARN

TODAY'S MINI-PROBLEMS

WRITE THE SYNTAX FROM MEMORY

MINI-PROBLEM 1

MINI-PROBLEM 2

OLD PROBLEM REVIEW

MISTAKE LOG

FINISH LINE

- ☐ I learned one syntax item.
- ☐ I solved two mini-problems.
- ☐ I reviewed one old problem.
- ☐ I wrote down what confused me.
- ☐ I stopped before overload.